DIY: 12-button macro pad for autodarts

Difficulty: Recommended for experienced manufacturers

Requires 3D printing and soldering

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List of materials:

- *12x GATERON G Pro 3.0 Switches: https://de.aliexpress.com/item/1005007052507996.html*
- *1x nRF52840 Developer Board (SuperMini): https://de.aliexpress.com/item/1005008664621573.html*
- *1x 503030 500mAh battery (also larger possible, as there is enough space): https://de.aliexpress.com/item/1005007964712378.html*
- *12x 1N4148 diodes: https://de.aliexpress.com/item/1005006208000285.html*
- *16x Jumper wires female (at least 20 cm → 30 cm recommended): https://de.aliexpress.com/item/1005007046465880.html*
- *1x Toggle switch (not necessary, but recommended for longer battery life in standby): https://de.aliexpress.com/item/4001252480288.html*
- *1x ASUS USB-BT500 Bluetooth 5.0 dongle adapter (not necessary if the device to be paired already has Bluetooth – but still recommended to save battery with BLE): https://www.cyberport.de/pc-und-zubehoer/netzwerk/bluetooth/asus/pdp/4e18-007/asus-usb-bt500-bluetooth-5-0-usb-adapter.html*

Required tools and additional materials:

- USB-C cable
- Solder
- Soldering iron
- Heat gun or lighter
- Heat Shrink Tubing
- Side Cutters
- Needle-nose pliers
- Stripping tool
- Superglue
- Filament – PLA or PETG or ... (2 different colors recommended)

3D printing:

Start by printing out all the models provided. If you're using a Bambu Lab printer, I recommend using the included .3mf files for better printing results.

To achieve a two-color print, a multi-color printer (with e.g. AMS). You can simply add a manual pause or a 'Filament Change' command to your slicer at the desired layer height. See: https://youtu.be/tKPWpWb8gBw. The printer will stop at the specified location, so you can swap the filament and resume printing.

To ensure the best print quality, you should not fill the printer with all models at the same time. I recommend grouping the prints as follows:

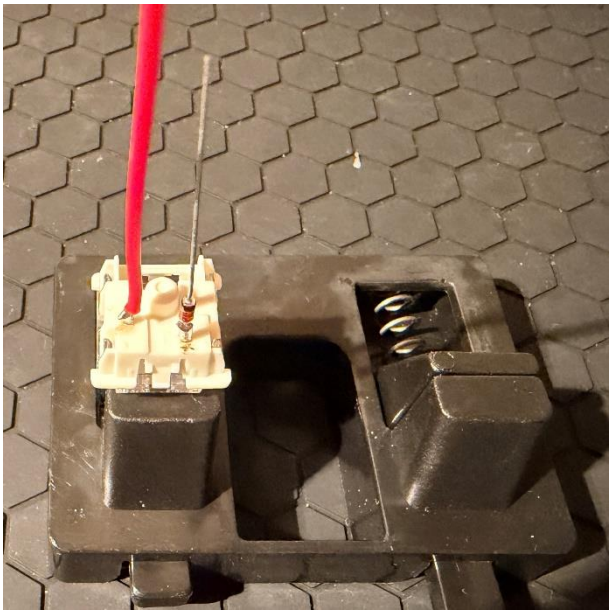
- Charge 1: Main case and the back
- Charge 2: All 12 keycaps (can be printed together)
- Charge 3: The optional wall mount should also be printed independently

Files on Printables: https://www.printables.com/model/1576041-autodarts-macropad

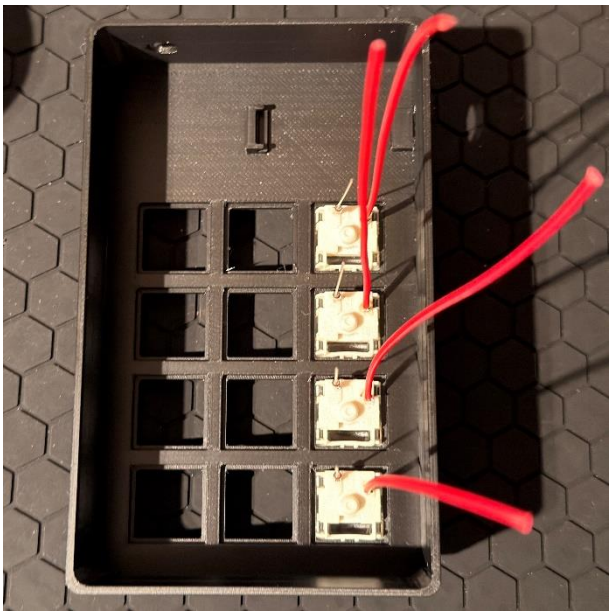
Files on MakerWorld: https://makerworld.com/de/models/2319788-autodarts-macropad

Soldering and assembly:

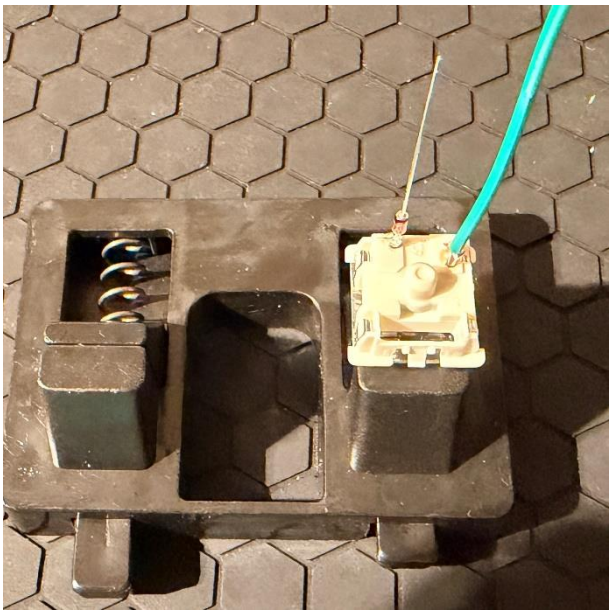
1. For the first four buttons, start by soldering a 15 cm long red wire to one of the contact pins. A 1N4148 diode is soldered to the opposite pin.
   CRITICAL: Align the diode so that the black line points away from the key.



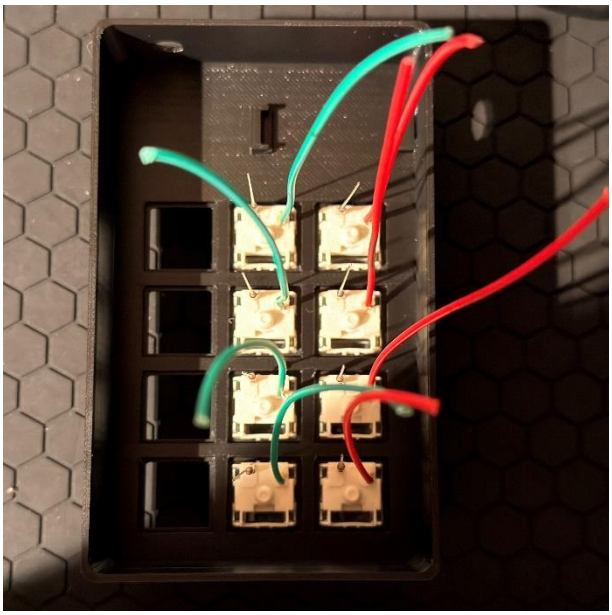
2. Snap the four pre-soldered keys into the housing vertically to form the first column. Make sure each key is tight.



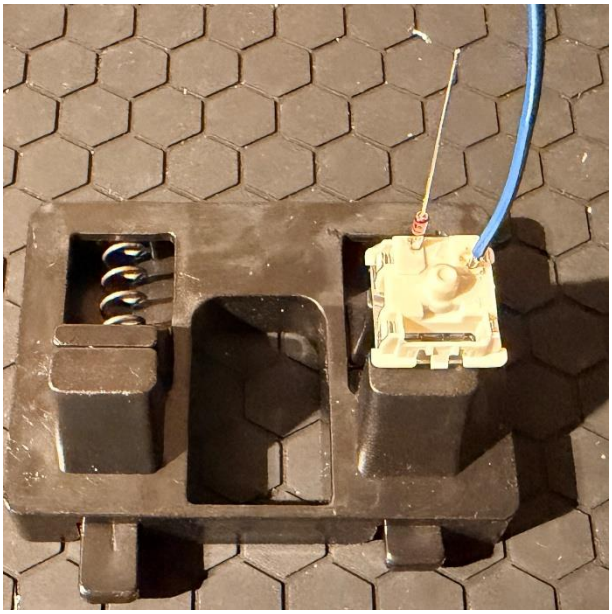
3. For the next four buttons, start by soldering a 15 cm long green wire to one of the contact pins. A 1N4148 diode is soldered to the opposite pin.
   CRITICAL: Align the diode so that the black line points away from the key.



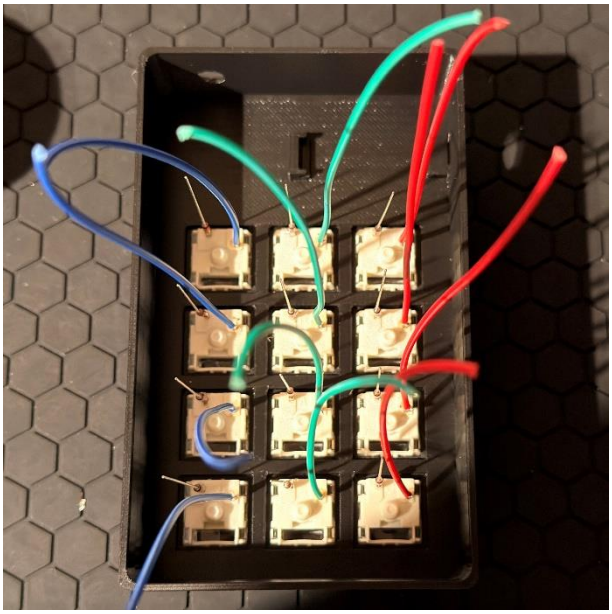
4. Snap the four pre-soldered buttons into the housing vertically to form the middle column. Make sure each key is tight.



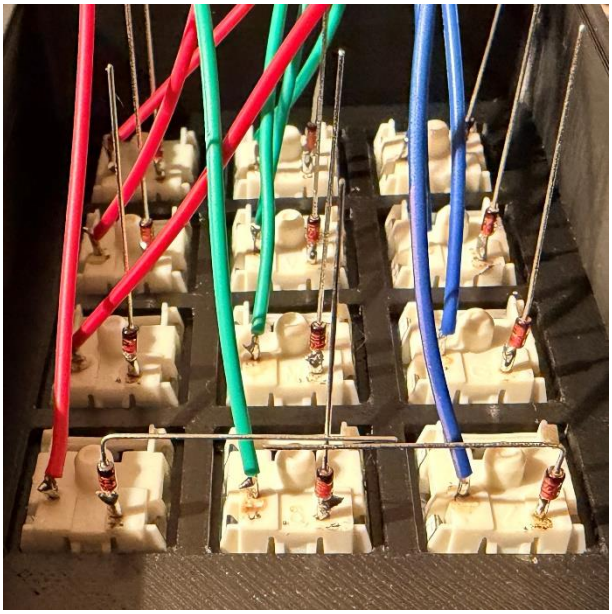
5. For the last four keys, start by soldering a 15 cm blue wire to one of the contact pins. A 1N4148 diode is soldered to the opposite pin.
   CRITICAL: Align the diode so that the black line points away from the key.



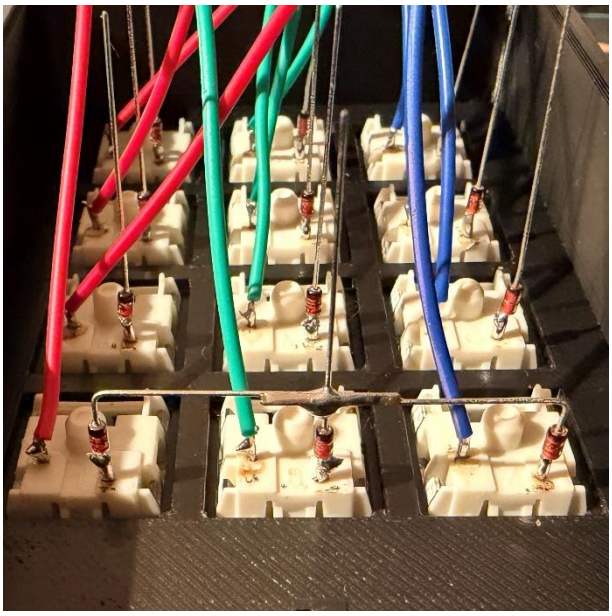
6. Snap the four pre-soldered buttons into the housing vertically to form the last column. Make sure each key is tight.



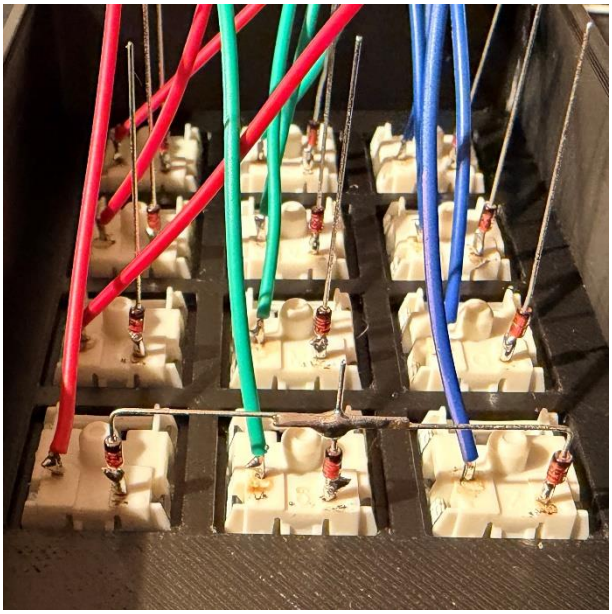
7. Identify the top row, indicated by the extra space reserved for the nRF52840. Bridge the three diodes by bending the outer terminals so that they overlap at the center diode. Using needle-nose pliers for this process ensures a more reliable bend.



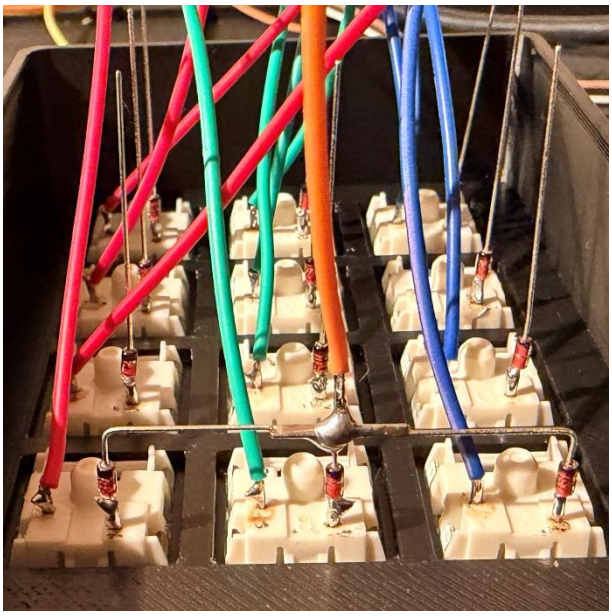
8. Now solder the connection where the three cathode lines meet. Make sure the connection is clean and covers all three lines to create a reliable rowbus.



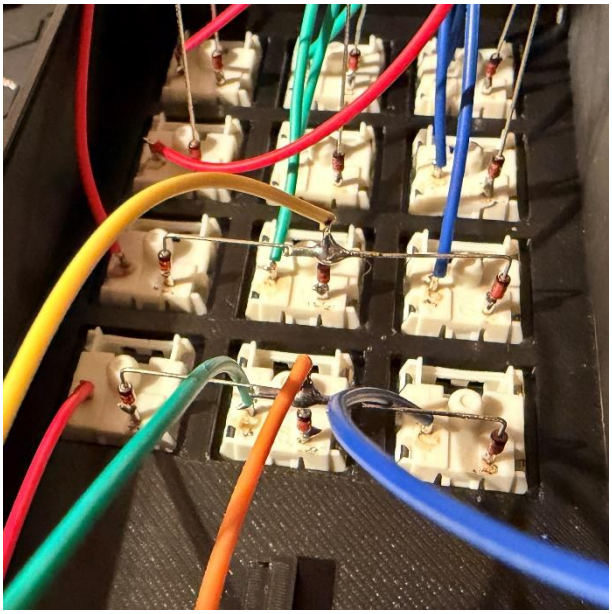
9. Use a side cutter to shorten the wire of the middle diode to about 8 mm in length. This leaves enough material for a secure connection for the next cable.



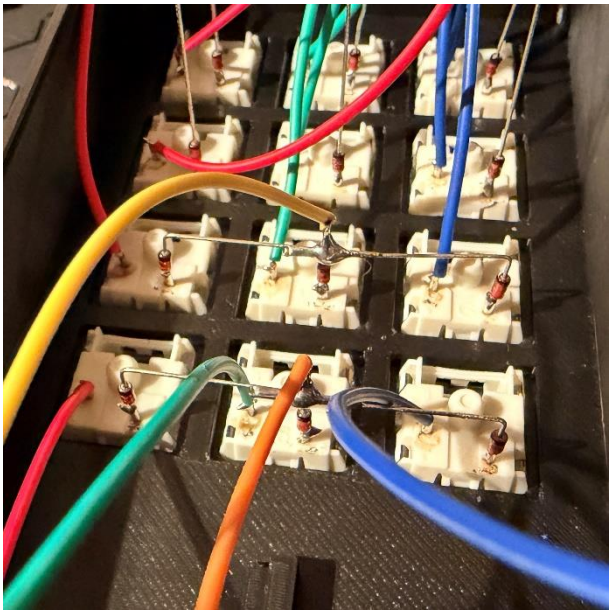
10. Solder a 20 cm orange cable with a female connector to the trimmed end of the diode. This cable serves as the main connection for your vertical row.



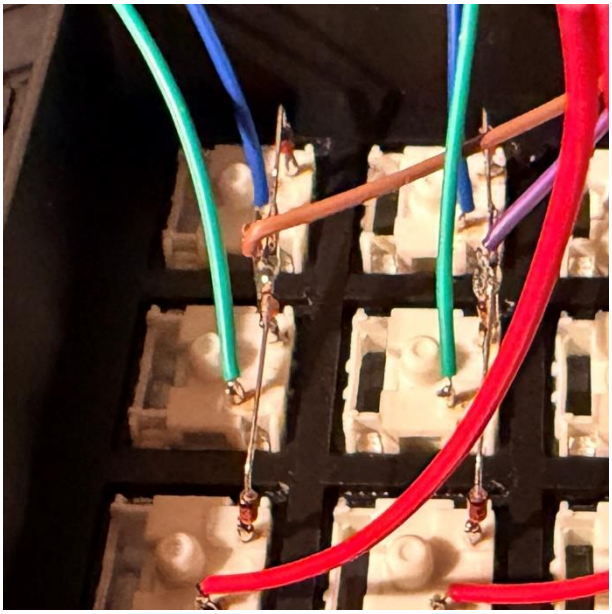
11. Repeat steps 7 through 10 for the second vertical column. However, replace the orange wire with a yellow wire. This color discrimination helps to correctly identify the rows during the final wiring phase.



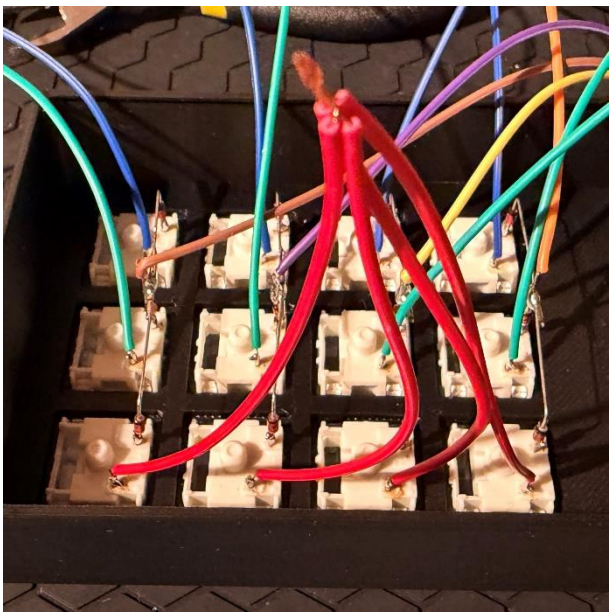
12. Repeat steps 7 through 10 for the second vertical column. However, replace the orange wire with a purple wire. This color discrimination helps to correctly identify the rows during the final wiring phase.



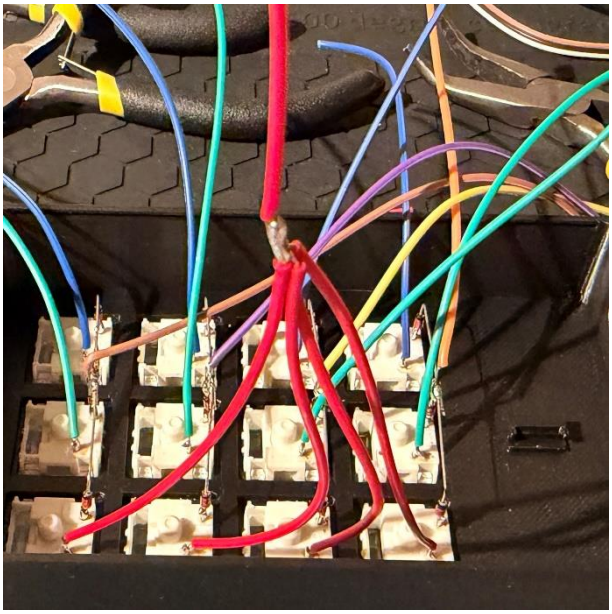
13. Repeat steps 7 through 10 for the second vertical column. However, replace the orange wire with a brown wire. This color discrimination helps to correctly identify the rows during the final wiring phase.



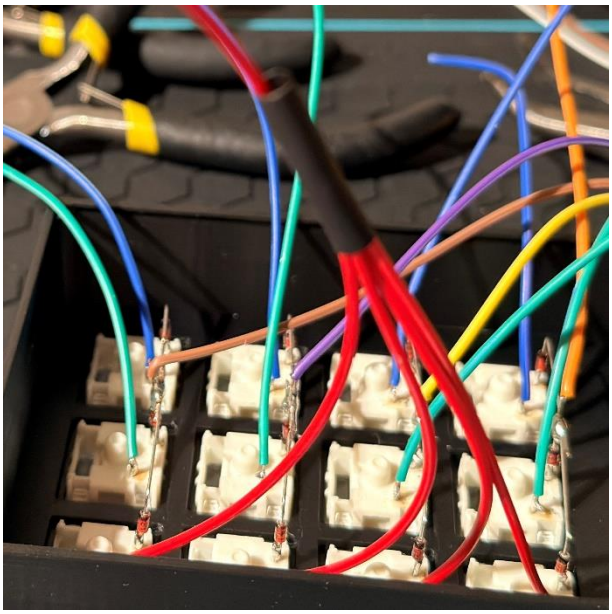
14. Strip off all the red wires and twist the ends of the copper strands together. Solder the bundle to create a single, secure termination point.



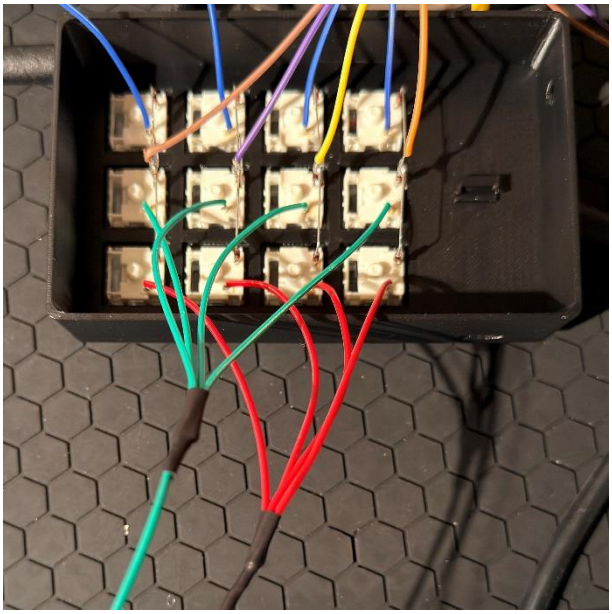
15. Prepare a 15 cm red cable with a female plug. Before soldering, slide a piece of heat shrink tubing onto the wire. Solder the open end of this cable to the red wire bundle you soldered in the previous step.



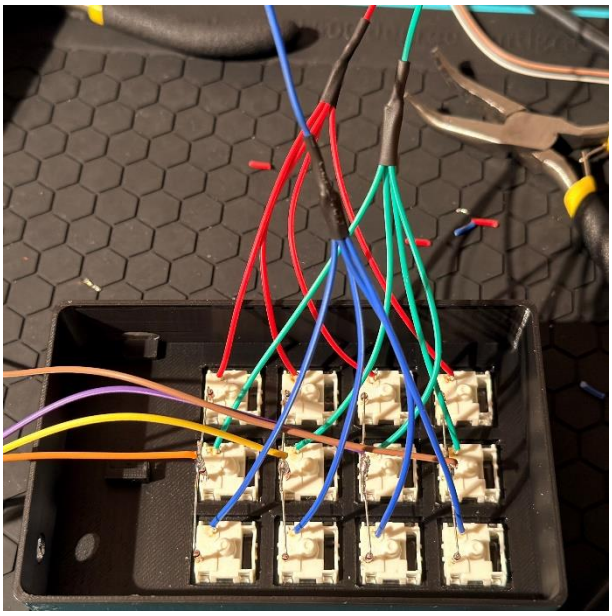
16. Slide the heat shrink tubing over the solder joint. Use a heat gun or lighter to shrink the hose until it forms a firm, secure seal around the joint.



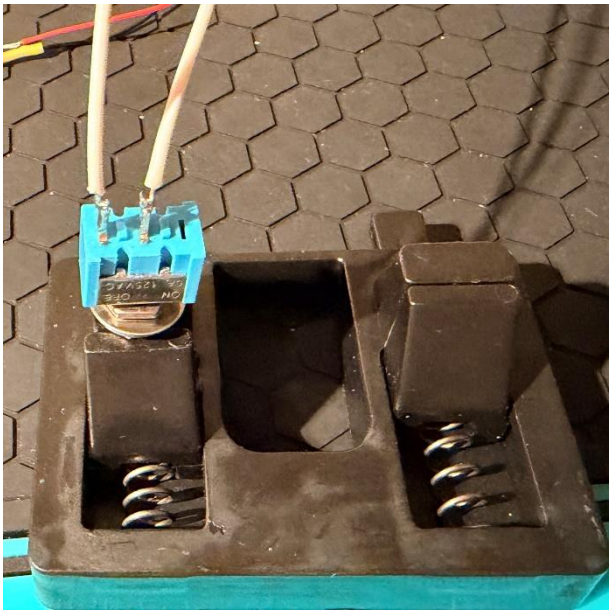
17. Repeat steps 14-16 with green wire instead of red for the middle column. Make sure that the solder joint is well insulated with heat shrink tubing as described earlier.



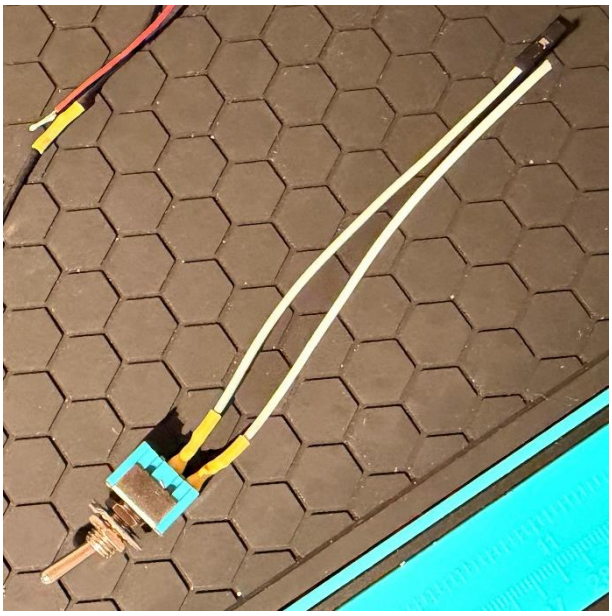
18. Repeat steps 14-16 with blue wire instead of red for the middle column. Make sure that the solder joint is well insulated with heat shrink tubing as described earlier.



19. Prepare the toggle switch by cutting a 30 cm female cable in half. Remove a few millimeters of insulation from the cut ends and slide a piece of heat shrink tubing onto each wire. Solder the two wires to the toggle switch terminals.

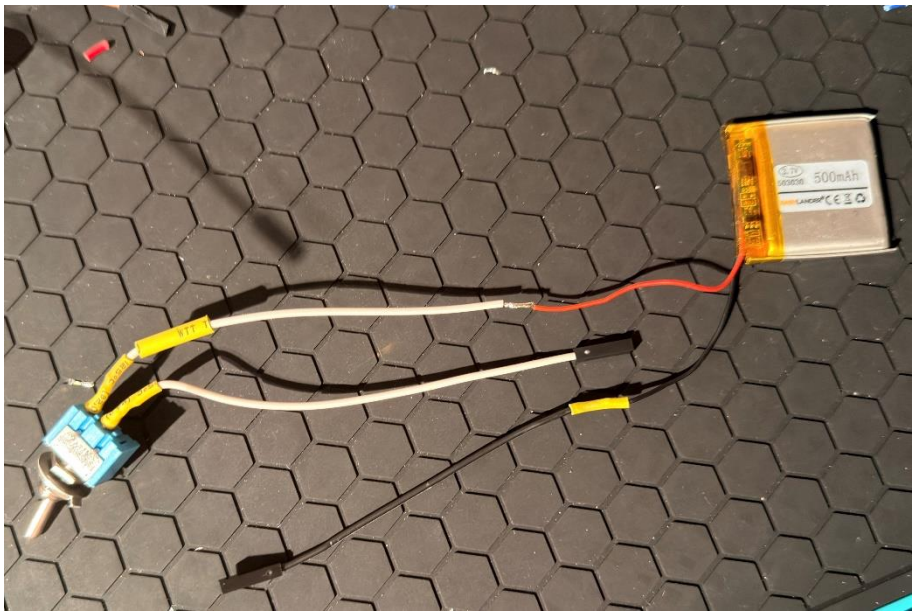


20. Slide the hose over the solder joints and shrink them with a heat gun or lighter for a fully insulated connection.

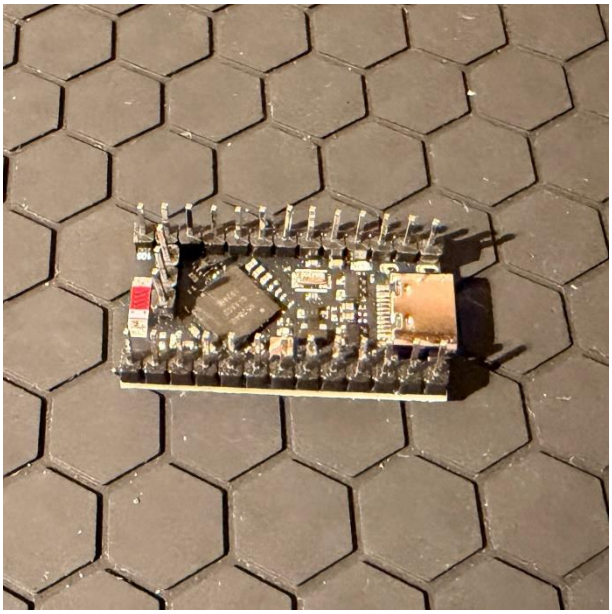


21.

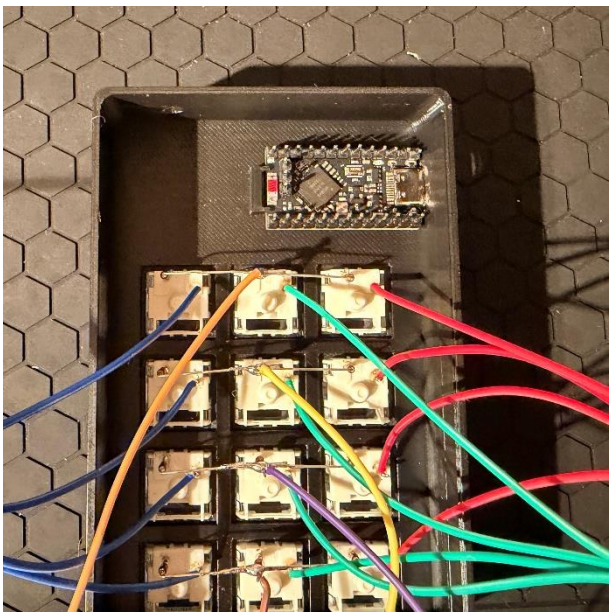
a. *Battery preparation:* If your battery has a plug, cut it off. WARNING: Cut the wires one at a time to prevent a short circuit. Don't let the exposed ends touch each other!
b. *Connect the toggle switch:* Strip the ends of the red battery cable and the toggle switch cable. Slide a piece of heat shrink tubing onto the red (positive) cable of the battery, then solder it to the toggle switch wires.
c. *Battery Ground Extension:* Take a 15cm black female jumper cable. Remove the insulation at the end and do the same with the negative (black) line of the battery. Slide heat shrink tubing onto the cable before soldering. Connect the two black wires and solder them together to make a solid connection.
d. *Insulation:* Position both heat shrink tubing over the solder joints and shrink them with hot air or a lighter to completely insulate the joints.



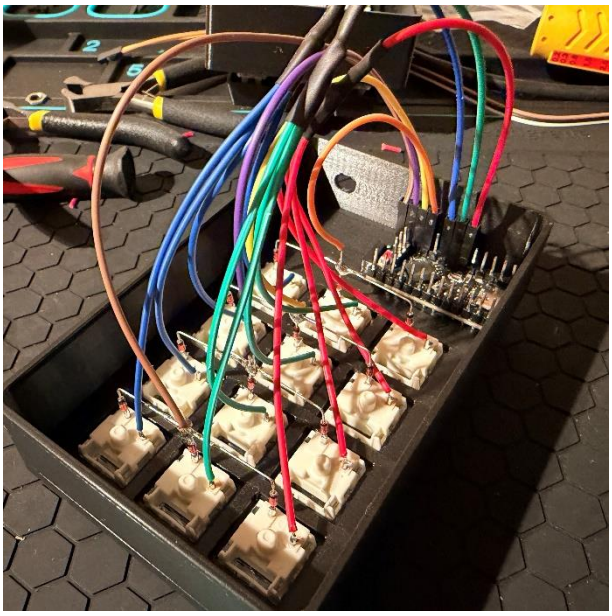
22. Unpack the microcontroller and solder the included male header pins to the board. Make sure the pins are perfectly vertical and the solder joints are clean, as these are the connection points for all of your female wires.



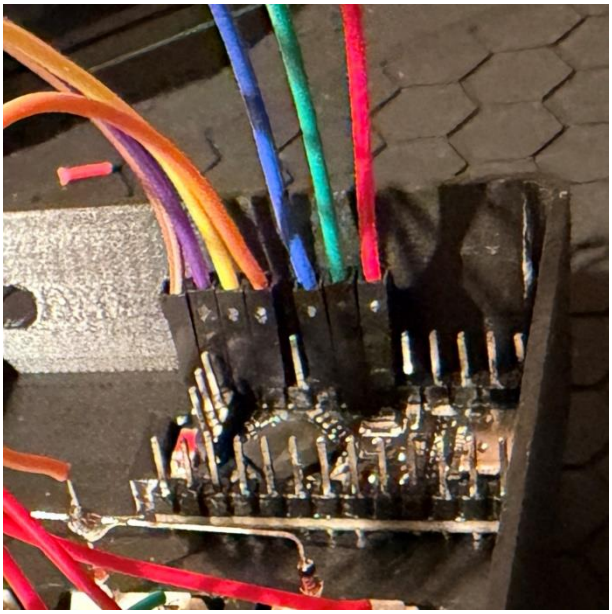
23. Place the controller in the case by aligning it with the two locking clips in the center of the case. You'll need to apply strong pressure near the USB-C port to slide the board down until the connector aligns perfectly with the case opening.
Note: You may need to move the controller carefully to find the perfect position. Be careful not to use excessive force so as not to damage the MCU or the 3D printed mounting tabs.

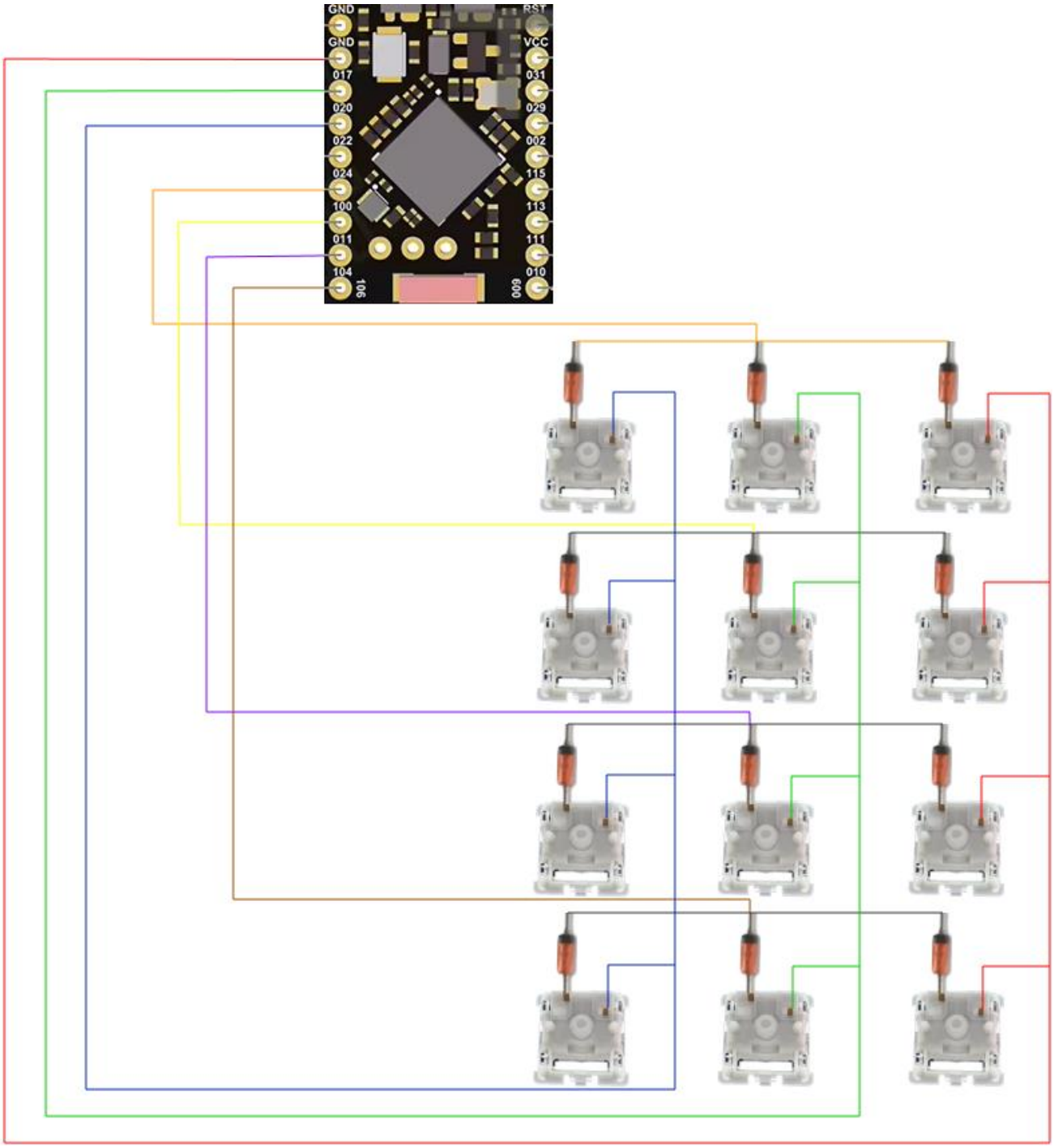


24. Now connect the female jumpers to the controller's header pins according to the schematic and pinout images provided.

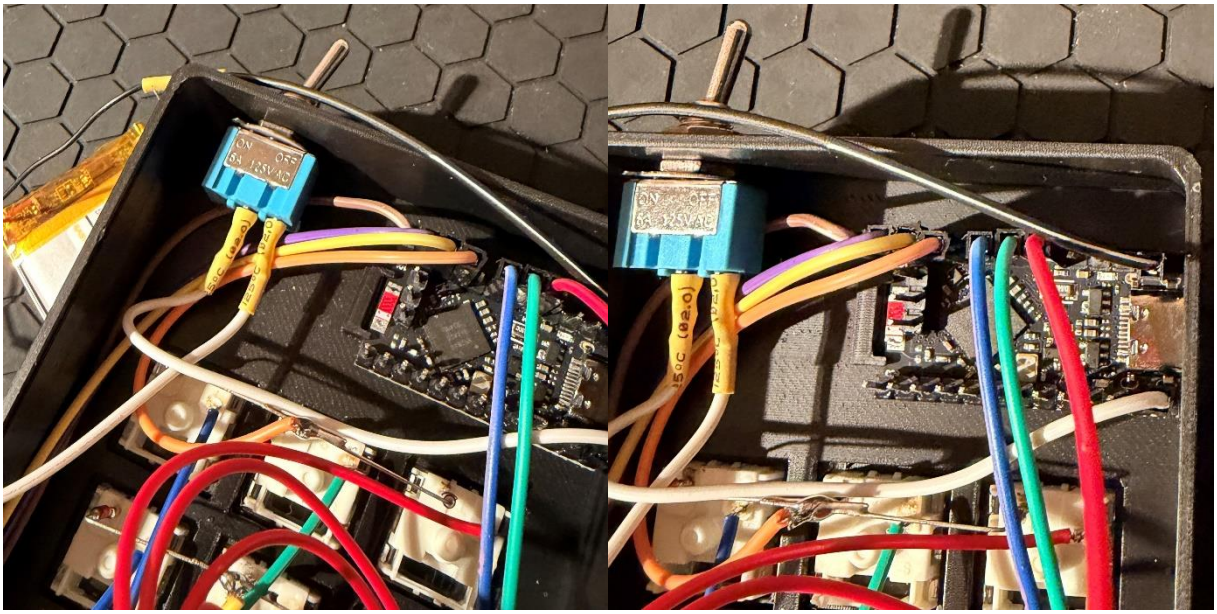


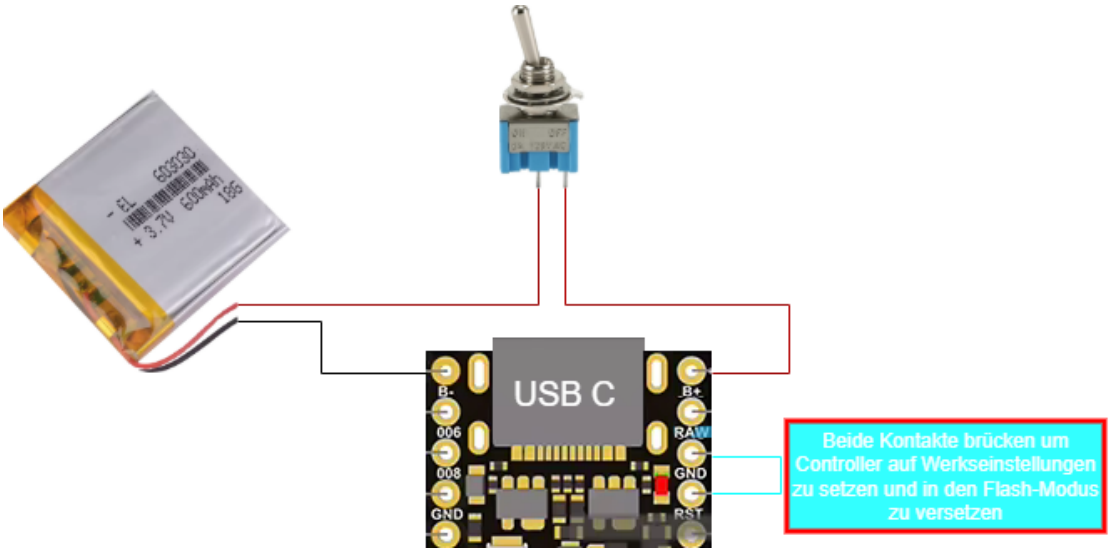
Important: Double-check that each row and column line is connected to its assigned GPIO pin. A single reversed wire will result in incorrect button presses or an unresponsive matrix.




GND
GND
017
020
022
024
100
011
104
106
RST
VCC
031
029
002
115
113
111
010
009

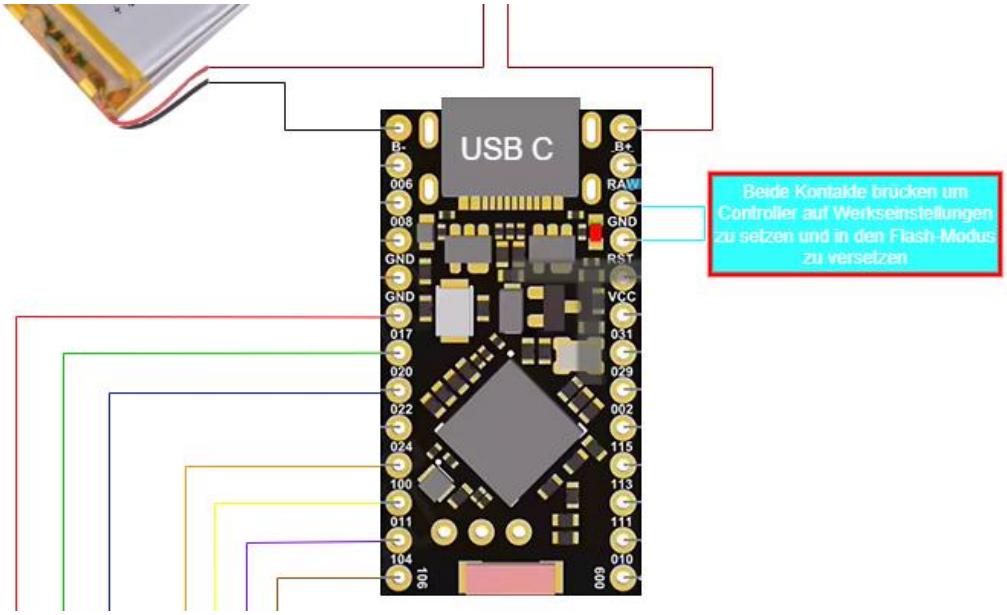
25. Install the toggle switch in the mounting hole provided in the case. Connect the female terminals (black and white) to the battery pads (B+ and B-) of the microcontroller, as shown in the reference image.
Important: Check the polarity before connecting. Swapping the battery lines may permanently damage the charging circuit of nRF52840.





26.

a. *Internal performance test:* Toggle the toggle switch to ON. The onboard status LEDs should light up immediately.
b. *USB connection:* Turn off the switch and connect the controller to your computer via USB-C. The LEDs will come back on, and your PC should recognize the device as a drive (e.g., NICENANO). **Troubleshooting:** > If the drive doesn't appear, you may need to manually trigger bootloader mode: Connect the USB cable and make sure the power switch is OFF. Use a screwdriver or tweezers to gently bridge the GND and RST pins twice in a row.
   <u>CAUTION:</u> Be extremely precise. Bridging the wrong pins could cause a short circuit and permanently damage the MCU.



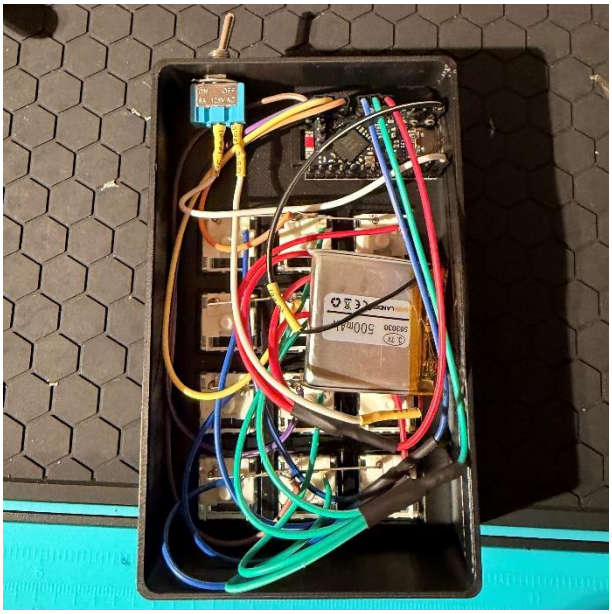
27.

a. *Install firmware:* Simply drag the file 6_Macropad_Firmware.uf2 (download here: https://github.com/mrcl-rckl/autodarts_macropad/releases/download/Autodarts/6_Macropad_Firmware.uf2) onto the controller's drive.
b. *Auto-restart:* The MCU automatically flashes the firmware and reboots. The drive will disconnect and disappear from your computer. Note: To reflash in the future, you will need to manually switch back to bootloader mode by bridging GND and RST pins twice in a row.
c. *Bluetooth pairing:* Unplug the USB-C cable and turn the power switch to ON. Your macropad is now recognizable as a Bluetooth keyboard in the settings of your autodarts system.

28.

a. *Power management note:* To preserve battery life, make sure that the device is always turned off via the toggle switch when it is not in use.
b. *Charging instructions:* To charge the battery, the toggle switch must be in the ON position. When the switch is off, the charging circuit is disconnected and the battery does not receive power from the USB-C port.
c. *Battery Life Efficiency:* Thanks to the low power consumption of the nRF52840, the battery is designed to last about 6 weeks on a single charge under normal usage conditions.

29. Carefully place the battery in its designated compartment in the case. Make sure all wires are neatly stowed away to avoid getting trapped. Finally, align the back cover and click it in to complete the Autodarts Macropad.



30. Firmly place the 12 keycaps on the switches by pressing them firmly on the keys. Make sure you follow the correct layout order that matches your firmware layout and Autodarts interface.

31. Last but not least: Attach the wall bracket to the wall with the appropriate screws. Once the holder is attached, simply slide the macropad into the holder until it is firmly in place and ready to use.

Test:

See the following image for the specific key assignments. To check that each key is working correctly and is mapped with the correct command, navigate to the online keyboard tester and press each key on your new macropad: https://www.onlinemictest.com/de/tastatur-test/

Einstellungen
werden geöffnet über gleichzeitigen Doppelklick von &
Mit Bluetooth Gerät 1 verbinden
Mit Bluetooth Gerät 2 verbinden
Mit Bluetooth Gerät 3 verbinden
Aktuell gewähltes Bluetooth Gerät verwerfen
Alle Bluetooth Geräte verwerfen
Pad neu starten
nicht belegt
nicht belegt
nicht belegt
1
2
3
Einstellungen verlassen
Pad in Update-Modus versetzen
Einstellungen verlassen

Schnell-Korrektur Menü: Tools for Autodarts benötigt
Belegung variiert nach Position des Darts, mögliche Korrekturen werden am Bildschirm angezeigt
S5
S20
S1
D5
D20
D1
T5
T20
T1
MISS
1
25
2
BULL
3

Operational Guide:

1. Turning the device on and off: Use the physical toggle switch on the side/back of the case. Power Saving: To maximize battery life, it is recommended to turn off the device when not in use, although the nRF52840 offers a power saving mode in deep sleep mode.

2. Battery level monitoring: The Autodarts Macropad reports its battery level directly to your operating system via the Bluetooth HID profile. You don't have to rely on internal LEDs. On Windows (11/10): Navigate to Settings > Bluetooth and Devices. Find the "Autodarts Macropad" in the list. The battery percentage is displayed right next to the device name. On Linux (GNOME/KDE): Open your System Preferences and go to the Power or Bluetooth section. The battery status is displayed under "Devices" or "Input Devices". Command Line (Advanced): You can also use the **upower -e command** to find the device path, and **upower -i [path]** to see the exact percentage.

3. Requirement for charging instructions: The toggle switch must be in the ON position to charge the battery. When the switch is off, the charging circuit is physically disconnected and the battery does not charge via USB-C.

4. Bluetooth pairing and resetting: Since the Macropad can store up to three different devices (e.g. PC, laptop, tablet), connecting and switching is done via the hidden admin level.
   To enable admin mode: Make sure the device is ON. Press the [Backspace] and [Enter] keys at the same time twice in quick succession (like a double-click). The pad will now switch to admin mode for 5 seconds. Switch Bluetooth device (Profile 1, 2, or 3): Turn on Admin Mode and press one of the top three buttons (row 1) to switch to Profile 1, 2, or 3. The pad will then automatically connect to the device stored there.
   Pair new PC (reset profile): If you want to connect the Macropad to a new computer, you must first delete the currently selected profile to trigger the pairing mode: Activate the admin mode (double-click on Backspace + Enter).

Press the left button in the second row. The device clears its memory for this profile and immediately switches to pairing mode. It will now be discoverable again as a new device in your system's Bluetooth settings.

Frequently Asked Questions:

Connectivity & Bluetooth:

Q: The connection feels delayed or the buttons repeat.

A: Bluetooth signals can be affected by interference. Make sure there are no large metal objects or powerful 2.4GHz routers directly between the pad and your computer. If you're using a desktop PC, make sure your Bluetooth antenna is plugged in.

Power & Battery:

Q: I've been plugging the pad into USB-C for hours, but the battery is still dead.

A: Crucial control: Is your toggle switch ON? Due to the hardware design, the battery only connects to the charging circuit when the switch is in the ON position. It does not charge when the device is turned off.

Q: The device works via USB, but it turns off immediately as soon as I unplug the cable.

A: Check the solder joints on the B+ and B pins of the microcontroller, as well as on the battery wires. Also, make sure that the wiring of the toggle switch is correct. If a cable has come loose, the battery power supply cannot reach the MCU.

Key Inputs & Hardware:

Q: A certain key is not registered at all.

A: This is usually a hardware problem. Check the solder joints on this exact key. Check the orientation of the diode. If the diode is soldered backwards, the signal will be blocked. Check the connection between the line/column cable and the MCU pin.

Q: When I press a key, the wrong letter appears on the screen.

A: Your physical cabling probably doesn't match the GPIO pins defined in the firmware. Check your wiring again with the wiring diagram on page 20. If you've swapped a row or column cable, the layout will be mirrored or moved.

Firmware & Flashing:

Q: The controller does not appear as "NICENANO" or "Removable Drive" when plugged in.

A: Use a different USB-C cable; Many cables are only on charging cables and do not transmit data. If the cable is fine, you'll need to manually enable bootloader mode.

Q: I flashed the .uf2 file, but nothing has changed.

A: Make sure the file transfer is complete before the drive is disconnected. If the drive disappears immediately after deleting the file, the flash was probably successful. If necessary, the Hard-Reset.uf2 file can be flashed first, which resets the controller. Afterwards, the normal firmware should be flashed again.

Car darts details:

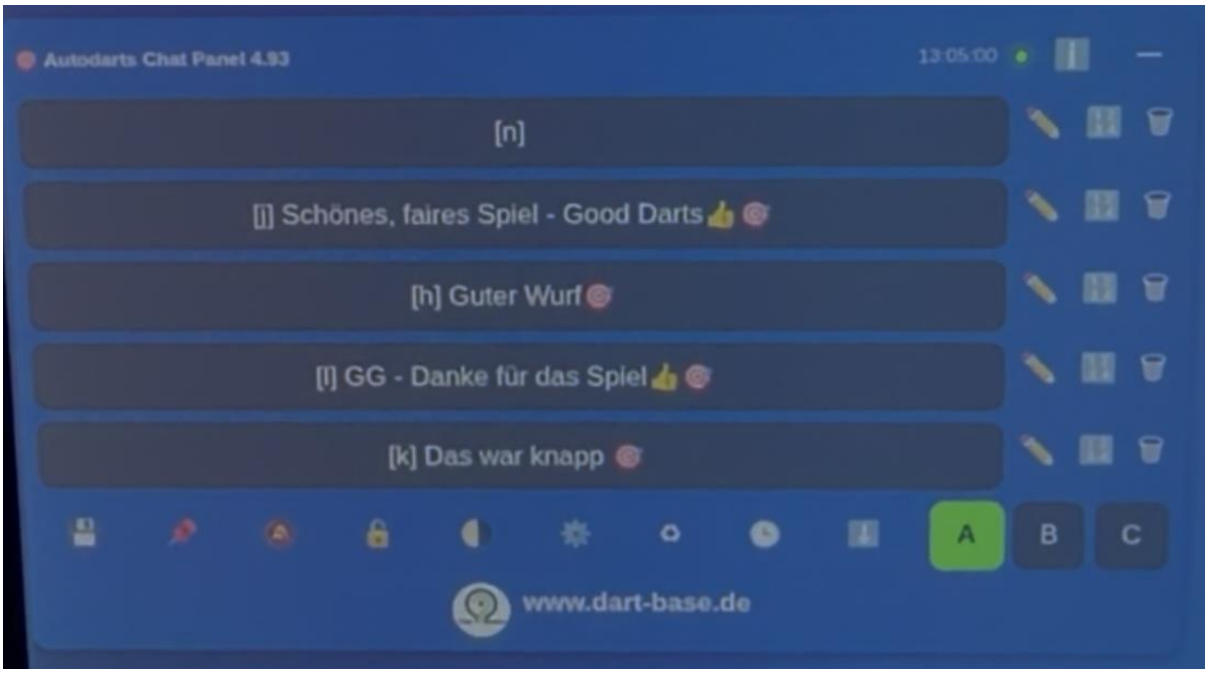
Q: No buttons work within the autodarts interface.

A: Make sure the browser window is "focused" (click in the Autodarts browser window).

Quick Chat Feature:

Q: How do I use the Quick Chat Macros?

A: Make sure that this script "Autodarts Chat Macro" is installed in Tampermonkey (download for Chrome here). Assign the macros to the H, J, K, and L keys.



Is your problem still not solved? If you've reviewed all of the above steps and the pad is still not working, please visit the GitHub Repository Issues page to report your issue or check for community-contributed solutions!

Firmware customization:

The Autodarts Macropad is fully customizable. If you want to change the key assignment or add new macros, you can adjust the ZMK firmware. Since the project is hosted on GitHub, you don't need to install any programming tools on your PC – the firmware is automatically created in the cloud.

The configuration of your buttons is stored in the "keymap" file. The original file can be found at this address: https://github.com/mrcl-rckl/autodarts_macropad/blob/main/config/macro12.keymap

How to change the layout:

- Fork the repository: Go to the main project page: https://github.com/mrcl-rckl/autodarts_macropad. Click the "Fork" button in the top right corner to create your own copy of the project.
- Edit the keymap: In your forked repository, go to the config/macro12.keymap file and click on the pencil icon to edit the code directly in your browser.
- Change keycodes: Replace the existing commands (e.g. &kp ENTER) by the desired commands.
    - *Example:* To change a mapping to the letter "M", use &kp M.
- Commit & Build: Click on "Commit changes" to save. This will automatically start a build process in the Actions tab of your GitHub repository.
- Download & Flash: Wait about 2-3 minutes. Once the build is complete (marked by a green checkmark), click on it, scroll down and download the firmware. Unzip the file and flash the .uf2 file as described on page 23.

Useful resources:

- ZMK Keycode Reference: A complete list of all available keys (media keys, F-keys, etc.) can be found here: https://zmk.dev/docs/codes

Thank you for your donation:

https://buymeacoffee.com/mrcl_rckl

If this DIY project hit the mark with you and you liked the building, you are welcome to support my work. Every donation is very welcome and helps me to develop more open source tools for the community.